

Part 2 of 2

The Industrial Opportunity for POWER OVER ETHERNET

This concluding part of the article discusses the market drivers for industrial Ethernet and how power over Ethernet (PoE) can meet the expanding needs

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Great strides have been made in promoting Ethernet use for industrial networks and factory automation. The challenge now is to further enhance this technology so that it becomes the leading industrial networking standard.

Introducing PoE in industrial applications creates new opportunities to directly power industrial equipment, drastically reduces overall installation and maintenance costs, improves overall reliability and increases configuration flexibility. Combining power and data over the same medium allows industrial Ethernet to compete with fieldbus.

This article discusses the market drivers for industrial Ethernet and how PoE can meet the expanding needs.

Ethernet migration from enterprise to the factory

Ethernet is a very popular communication standard widely deployed in enterprise applications. Driven by IP convergence, it is migrating from the enterprise level to the factory floor in the form of industrial Ethernet.

Adopting a common standard facilitates integration and communication between machines and equipment. Sharing data from the machine level to the enterprise resource planner platform becomes possible over a single integrated network. This integrated network—where data

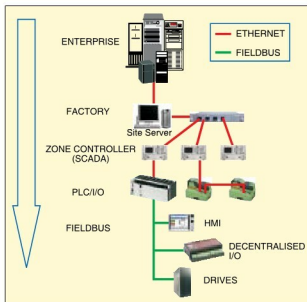


Fig. 1: Industrial network migration has reached the PLC level

on every machine can be accessed easily—simplifies logistic process control, including tracking inventory, work-in-process control and production planning.

Ethernet everywhere has many advantages. For example, when reconfiguring a production system, a common communication system supports adaptations and extensions in a flexible way. Adding modules is plug-and-play. Security and reliability are improved with the common Ethernet standard, which is well proven and robust. Installation and replacement costs with a standard are significantly lower than with a proprietary interface. Furthermore, Ethernet enjoys widespread usage and is available worldwide, which means that industrial Ethernet can

take advantage of COTS technology and equipment.

Initially, Ethernet connectivity appeared in industrial devices relatively high in the industrial network hierarchy and closer to the enterprise network. The hierarchy starts with the enterprise server network and expands toward the factory server and zone controllers. Each zone has programmable logic controller (PLC) devices controlling several sub-modules. Today, migration has reached the PLC level

as shown in Fig. 1.

Worldwide shipments of industrial input/output (I/O) modules with Ethernet-enabled connectivity comprised 14.7 per cent of the total market in 2005 and are expected to account for 20.6 per cent of the total by 2010.

Industrial Ethernet protocols and competitors

The Ethernet standard addresses physical and data link layers of communication protocols. These layers are at the bottom of seven-layer OSI model as shown in Fig. 2. The Ethernet protocol defines the Ethernet 10/100BASE-T physical media over which data is sent and the CSMA/CD access protocol for different devices connected to the media.